

Basic Electrical Engineering

MODULE 1 — PART 2: QUESTION BANK

Module 1 (10 hrs): DC Circuits

For First-Year Students — with complete answers and diagrams

Course: Basic Electrical Engineering · Course Code: EE100

Question Paper Structure

Section	Marks per Question	Scope	Questions
Section A	2 marks	Compulsory — from every module	Q1 - Q32
Section B	5 marks	From Modules 1 and 2 only	Q1 - Q14
Section C	7 marks	Compulsory — from every module	Q1 - Q9
Section D	Numericals	Solved problems (all topics)	N1 - N12

- Every section covers **all the topics** of Module 1 — no topic is left out.
- No question is repeated between sections.
- Answer 2-mark questions in 2–4 lines, 5-mark questions with a diagram in about half a page, and 7-mark questions with full explanation and diagrams.
- Section D gives every numerical problem solved step by step, with its circuit diagram.

SECTION A — 2-Mark Questions

Q1. What is an electric circuit? Distinguish between active and passive elements.

Answer: An electric circuit is an interconnection of elements with at least one closed path in which current can flow. Active elements supply energy (voltage/current sources, generators); passive elements only receive energy (resistors, inductors, capacitors).

Q2. Define voltage and give its unit.

Answer: Voltage (potential difference) is the energy required to move a unit charge through an element. It is measured in volts (V): $v = dW/dq$.

Q3. Define electric current and give its unit.

Answer: Electric current is the time rate of change of charge, measured in amperes (A): $i = dq/dt$.

Q4. Define power and energy with their formulas.

Answer: Power is the rate of change of energy: $p = vi = i^2R = v^2/R$ (watts). Energy is the capacity to do work (joules): Energy = Power × Time. 1 Wh = 3600 J.

Q5. What is the passive sign convention?

Answer: It is satisfied when the current enters through the positive terminal of an element, giving $p = +vi$. If the current enters through the negative terminal, $p = -vi$.

Q6. Give the difference between an ideal and a practical voltage source.

Answer: An ideal voltage source maintains a constant terminal voltage regardless of the current drawn. A practical voltage source has a low but finite internal resistance, so its terminal voltage falls as the load current rises.

Q7. Give the difference between an ideal and a practical current source.

Answer: An ideal current source supplies the same current to any load. A practical current source has a high but finite internal resistance, so the load current changes when the load resistance changes.

Q8. What are dependent sources? Name their four types.

Answer: Sources whose voltage or current depends on another voltage or current in the circuit. Types: VCVS, CCVS, VCCS and CCCS.

Q9. Define resistance. On what factors does it depend?

Answer: Resistance is the property that opposes the flow of current, converting electrical energy into heat. It depends on length, cross-sectional area, type of material (resistivity) and temperature.

Q10. Write the formula for the resistance of a conductor.

Answer: $R = \rho l / a$, where ρ is the resistivity (ohm-metre), l the length and a the cross-sectional area.

Q11. What is an inductor? Write its voltage equation.

Answer: An inductor is an energy-storage device that stores energy in a magnetic field and opposes any change in current. $v_L = L di/dt$.

Q12. What is a capacitor? Write its current equation.

Answer: A capacitor is a charge-storage device made of two plates separated by a dielectric. $i_C = C dv/dt$.

Q13. What is an open circuit? What is a short circuit?

Answer: An open circuit has resistance approaching infinity (no current flows). A short circuit has resistance approaching zero (full current with no voltage drop).

Q14. State Ohm's law.

Answer: The current through a conductor is directly proportional to the potential difference across it, provided the physical conditions remain constant: $V = IR$.

Q15. What is a series circuit? Write the equivalent resistance.

Answer: Elements are in series when they carry the same current. The equivalent resistance is the sum of the individual resistances: $R_{eq} = R_1 + R_2 + \dots$

Q16. State the voltage division rule.

Answer: In a series circuit the voltage divides in proportion to the resistances: $V_1 = V R_1/(R_1 + R_2)$.

Q17. What is a parallel circuit? Write the equivalent resistance of two resistors in parallel.

Answer: Elements are in parallel when they have the same voltage across them. For two resistors, $R_{eq} = R_1 R_2 / (R_1 + R_2)$.

Q18. State the current division rule.

Answer: In a parallel circuit the current divides in inverse proportion to the resistances: $I_1 = I R_2 / (R_1 + R_2)$.

Q19. State Kirchhoff's Current Law (KCL).

Answer: The algebraic sum of the currents meeting at a junction is zero, i.e. the sum of the currents entering a node equals the sum of the currents leaving it. It is based on conservation of charge.

Q20. State Kirchhoff's Voltage Law (KVL).

Answer: In any closed circuit or mesh, the algebraic sum of all the emfs and the voltage drops in the resistors is zero. It is based on conservation of energy.

Q21. Define node, branch, loop and mesh.

Answer: Node: a junction where two or more elements meet. Branch: a section between two junctions. Loop: any closed path. Mesh: a closed path containing no other loop within it.

Q22. Write the delta-to-star conversion formulas.

Answer: Each star resistance = (product of the two adjacent delta resistances) / (sum of the three delta resistances), e.g. $R_1 = R_{AB}R_{CA}/(R_{AB}+R_{BC}+R_{CA})$.

Q23. Write the star-to-delta conversion formula.

Answer: Each delta resistance = (sum of the products of the star resistances taken in pairs) / (opposite star resistance), e.g. $R_{AB} = (R_1R_2+R_2R_3+R_3R_1)/R_3$.

Q24. What is mesh analysis? When is it useful?

Answer: Mesh analysis assigns a current to each mesh and writes KVL for each mesh in terms of the mesh currents. It is applicable to planar networks and is useful when the network has many voltage sources.

Q25. What is nodal analysis? When is it useful?

Answer: Nodal analysis chooses a reference node and writes KCL at each non-reference node in terms of node voltages. It is useful when the network has many current sources.

Q26. What is source transformation? Write the relation.

Answer: A voltage source with a series resistance is converted into an equivalent current source with a parallel resistance (and vice-versa): $I_S = V_S/R_S$ and $R_P = R_S$.

Q27. State the superposition theorem.

Answer: In a linear, bilateral network with more than one source, the response in any element equals the algebraic sum of the responses due to each source acting alone, with the other voltage sources shorted and current sources opened.

Q28. State Thevenin's theorem.

Answer: Any linear bilateral network with terminals A and B can be replaced by a single voltage source V_{Th} in series with a resistance R_{Th} , where V_{Th} is the open-circuit voltage across A-B and R_{Th} is the resistance seen from A-B with sources removed.

Q29. State Norton's theorem.

Answer: Any linear bilateral network with terminals A and B can be replaced by a single current source I_N in parallel with a resistance R_N , where I_N is the short-circuit current through A-B and $R_N = R_{Th}$.

Q30. State the maximum power transfer theorem.

Answer: In a DC circuit, maximum power is transferred from the source to the load when the load resistance equals the internal resistance of the source as seen from the load terminals ($R_L = R_i$).

Q31. Write the relation between the Thevenin and Norton equivalents.

Answer: $I_N = V_{Th} / R_{Th}$ and $R_N = R_{Th}$.

Q32. What is the efficiency at maximum power transfer?

Answer: The efficiency is only 50%, because half of the total power generated is dissipated in the internal resistance of the source.

SECTION B — 5-Mark Questions**Q1. Explain voltage, current, power and energy with their formulas.**

Answer:

- **Voltage:** energy needed to move a unit charge; $v = dW/dq$ (volts).
- **Current:** rate of flow of charge; $i = dq/dt$ (amperes).
- **Power:** rate of change of energy; $p = vi = i^2R = v^2/R$ (watts).
- **Energy:** capacity to do work; Energy = Power × Time (joules), and 1 Wh = 3600 J.

By the passive sign convention, $p = +vi$ when the current enters the positive terminal (power absorbed), and $p = -vi$ when it enters the negative terminal (power supplied).

Q2. Explain the different types of sources.

Answer:

- **Ideal voltage source:** constant terminal voltage regardless of the current drawn.
- **Practical voltage source:** has a low finite internal resistance, so its terminal voltage falls as load current rises.
- **Ideal current source:** supplies a constant current to any load.
- **Practical current source:** has a high finite internal resistance, so the load current changes with the load.
- **Dependent sources:** their value depends on another circuit voltage or current (VCVS, CCVS, VCCS, CCCS).

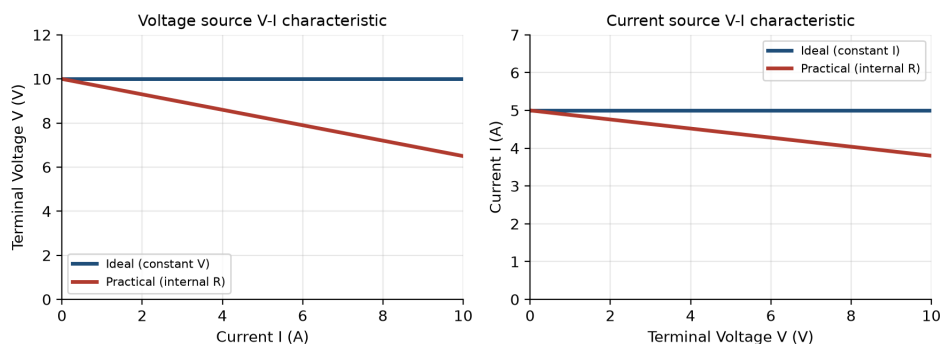


Figure 1: V-I characteristics of ideal and practical sources.

Q3. Explain resistance, inductance and capacitance.

Answer:

- **Resistance:** opposes current flow, converting electrical energy to heat; $V = IR$; $R = \rho l/a$; depends on length, area, material and temperature.
- **Inductance:** stores energy in a magnetic field and opposes any change in current; $v_L = L di/dt$; $L = \mu N^2 A/l$.
- **Capacitance:** stores charge in an electric field between two plates separated by a dielectric; $i_C = C dv/dt$.

Q4. Explain series and parallel circuits with the voltage and current division rules.

Answer:

- **Series:** same current through all elements; $R_{eq} = R_1 + R_2 + \dots$; voltage divides as $V_1 = V R_1 / (R_1 + R_2)$.
- **Parallel:** same voltage across all elements; $1/R_{eq} = 1/R_1 + 1/R_2 + \dots$; for two resistors $R_{eq} = R_1 R_2 / (R_1 + R_2)$; current divides as $I_1 = I R_2 / (R_1 + R_2)$.

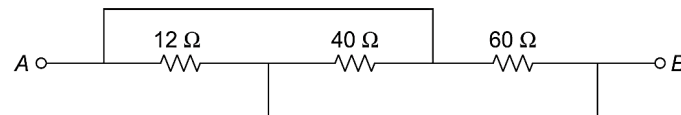


Figure 2: Series resistances.

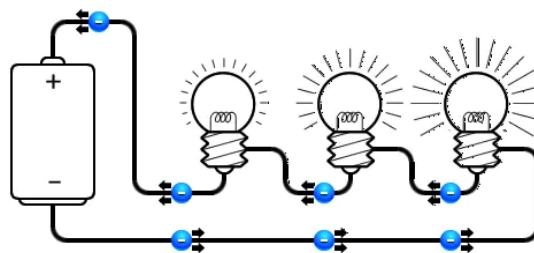


Figure 3: Parallel resistances.

Q5. State and explain Kirchhoff's Current Law and Voltage Law.

Answer:

KCL: the algebraic sum of the currents at a junction is zero — the sum of the currents entering a node equals the sum of the currents leaving it. It follows from conservation of charge.

KVL: in any closed loop, the algebraic sum of all the emfs and the voltage drops in the resistors is zero. It follows from conservation of energy.

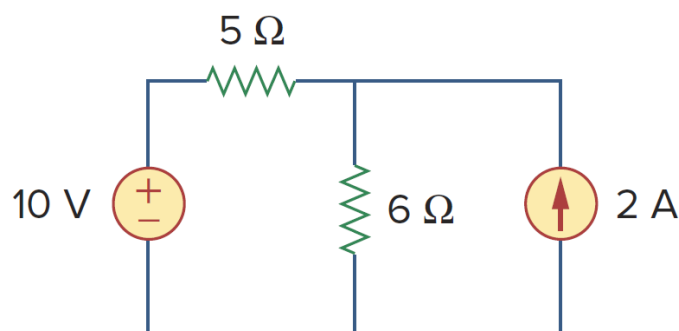


Figure 4: KCL at a junction.

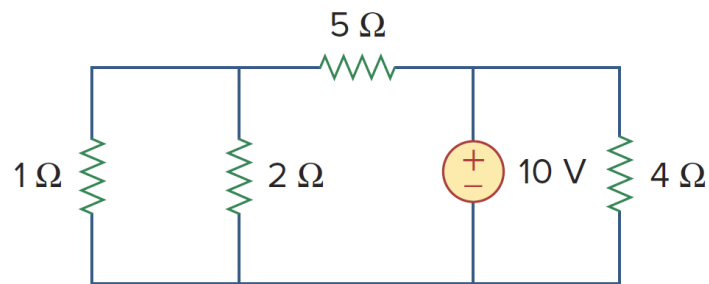


Figure 5: KVL around a loop.

Q6. Explain nodes, branches, loops and meshes with a diagram.**Answer:**

- **Node:** a junction where two or more elements meet.
- **Branch:** the section of a circuit between two junctions.
- **Loop:** any closed path in a circuit.
- **Mesh:** a closed loop that contains no other loop within it.

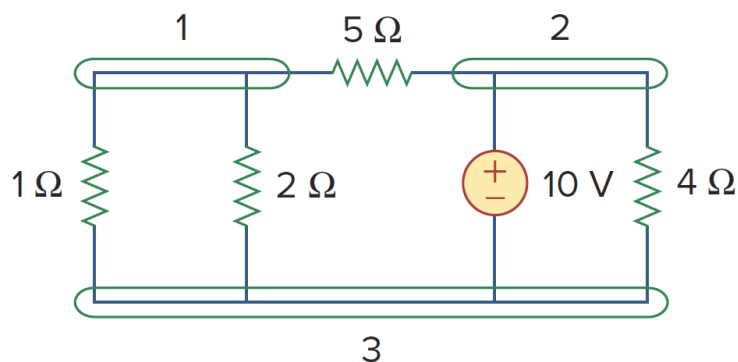


Figure 6: Nodes, branches, loops and meshes.

Q7. Explain star-delta and delta-star conversion with formulas.**Answer:**

A star (Y) network can be converted into an equivalent delta (Δ) network and vice-versa — very useful for simplifying networks that cannot be reduced by series/parallel combinations.

$$\Delta \rightarrow Y: R_1 = R_{AB}R_{CA}/(R_{AB}+R_{BC}+R_{CA})$$

$$Y \rightarrow \Delta: R_{AB} = (R_1R_2+R_2R_3+R_3R_1)/R_3$$

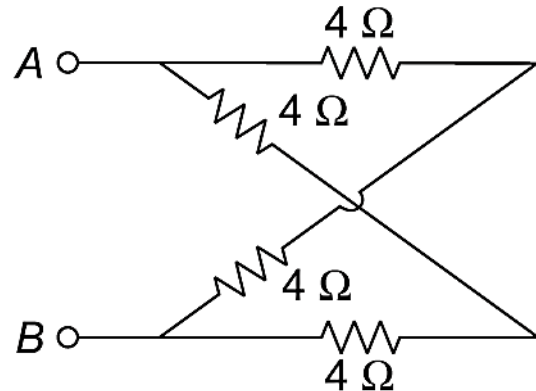


Figure 7: Star and delta networks.

Q8. Explain mesh analysis with its steps.**Answer:**

Mesh analysis assigns a current to each mesh and applies KVL around each mesh. It applies only to planar networks and is useful when there are many voltage sources.

- 1 Identify the meshes, assign a direction and an unknown mesh current to each.
- 2 Assign the polarities of the branch voltages.
- 3 Apply KVL around each mesh and express the branch voltages using Ohm's law.
- 4 Solve the simultaneous equations for the mesh currents.

	Series circuit	Parallel network
Current	The current is the same in all parts of the circuit $I = I_1 = I_2 = I_3$	The total current supplied to the network equals the sum of the currents in the various branches $I = I_1 + I_2 + I_3$
Voltage	The total voltage equals the sum of the voltages across the different parts of the circuit $V = V_1 + V_2 + V_3$	The voltage across a parallel combination is the same as the voltage across each branch $V = V_1 = V_2 = V_3$
Resistance	The total resistance equals the sum of the separate resistances $R = R_1 + R_2 + R_3$	The reciprocal of the equivalent resistance equals the sum of the reciprocals of the branch resistances $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

Figure 8: A multi-mesh network solved by mesh analysis.

Q9. Explain nodal analysis with its steps.**Answer:**

Nodal analysis chooses a reference node and writes KCL at each non-reference node in terms of the node voltages. It is useful when the network has many current sources.

- 1 Choose a reference (ground) node and label the other node voltages $V_1, V_2, \dots$
- 2 Assume all currents to flow away from the node under consideration.
- 3 Apply KCL at each non-reference node, writing each branch current as (node-voltage difference)/resistance.
- 4 Solve the equations for the node voltages, then find any branch current by Ohm's law.

Q10. Explain source transformation.

Answer:

A voltage source with a series resistance can be converted into an equivalent current source with a parallel resistance, and a current source with a parallel resistance can be converted into an equivalent voltage source with a series resistance.

$$I_S = V_S/R_S \quad \text{and} \quad R_P = R_S$$

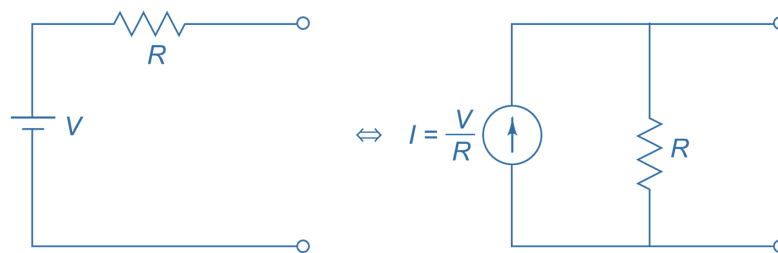


Figure 9: Source transformation.

Q11. State the superposition theorem and explain its procedure.

Answer:

In a linear, bilateral DC network with more than one source, the resultant voltage or current in any element equals the algebraic sum of the values produced by each source acting alone, with the other voltage sources short-circuited and current sources open-circuited.

- 1 Select one source; replace all other voltage sources by short circuits and current sources by open circuits.
- 2 Find the desired voltage/current due to this single source.
- 3 Repeat for each remaining source.
- 4 Add all the contributions algebraically.

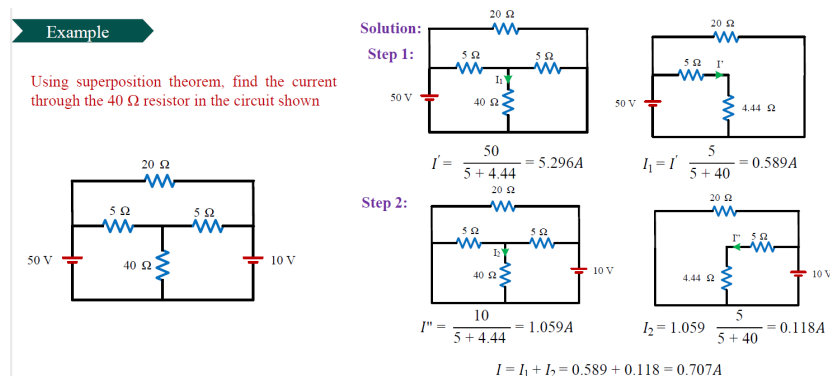


Figure 10: Superposition — worked example.

Q12. State Thevenin's theorem and explain the procedure to find the Thevenin equivalent.

Answer:

Any linear, bilateral network with terminals A and B can be replaced by a single voltage source V_{Th} in series with a resistance R_{Th} .

- 1 Remove the load between the two terminals.
- 2 Find the open-circuit voltage across the terminals — this is V_{Th} .
- 3 Find the resistance between the terminals with all voltage sources shorted and current sources opened — this is R_{Th} .
- 4 Connect V_{Th} and R_{Th} in series, reconnect the load, and use Ohm's law to find the load current.

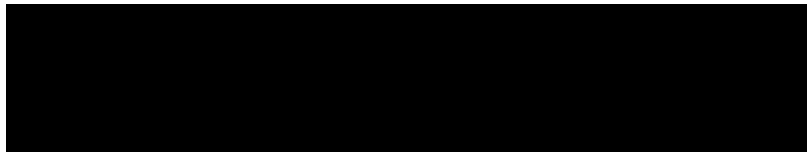


Figure 11: Thevenin's theorem.

Q13. State Norton's theorem and give its equivalent circuit.

Answer:

Any linear, bilateral network with terminals A and B can be replaced by a single current source I_N in parallel with a resistance R_N . I_N is the short-circuit current through A-B and R_N is found exactly like R_{Th} . The relation with the Thevenin equivalent is $I_N = V_{Th}/R_{Th}$ and $R_N = R_{Th}$.

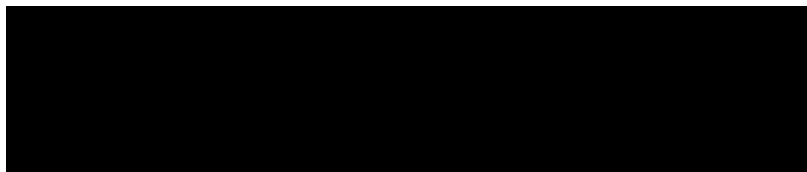


Figure 12: Norton's theorem.

Q14. State the maximum power transfer theorem and give the value of maximum power.

Answer:

In DC circuits, maximum power is transferred to the load when the load resistance equals the internal resistance of the source as seen from the load terminals: $R_L = R_i$. The maximum power is $P_{max} = V^2/(4R_L)$, and the efficiency at maximum power transfer is only 50%.

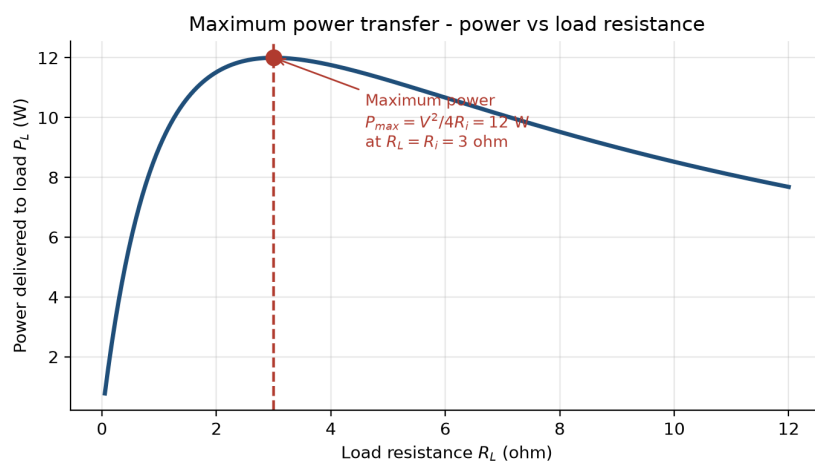


Figure 13: Power vs load resistance — peak at $R_L = R_i$

SECTION C — 7-Mark Questions

Q1. Explain the basic circuit elements and sources in detail (voltage, current, power, energy, resistance, inductance, capacitance and the types of sources).

Answer:

Circuit: an interconnection of elements with at least one closed path. Active elements supply energy; passive elements only receive energy.

- Voltage: $v = dW/dq$ (V). Current: $i = dq/dt$ (A). Power: $p = vi = i^2R = v^2/R$ (W). Energy = Power $\times$ Time (J); 1 Wh = 3600 J.
- Resistance: opposes current, $V = IR$, $R = \rho l/a$ (depends on length, area, material, temperature).
- Inductance: stores energy in a magnetic field, $v_L = L di/dt$.
- Capacitance: stores charge, $i_C = C dv/dt$.

Sources: an ideal voltage source holds a constant terminal voltage; a practical voltage source has a finite internal resistance so its voltage falls with load current. An ideal current source gives a constant current; a practical one has a finite internal resistance. Dependent (controlled) sources are of four types: VCVS, CCVS, VCCS and CCCS.

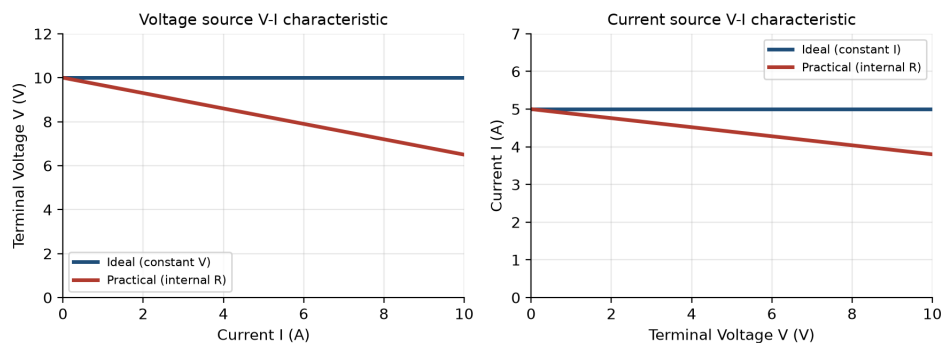


Figure 14: Ideal and practical source characteristics.

Q2. Explain series, parallel and series-parallel circuits with the voltage and current division rules.

Answer:

Series: elements carry the same current. The equivalent resistance is the sum of the individual resistances, $R_{eq} = R_1 + R_2 + R_3$. The voltage divides in proportion to the resistances: $V_1 = V R_1/(R_1+R_2)$.

Parallel: elements have the same voltage across them. The reciprocal of the equivalent resistance equals the sum of the reciprocals: $1/R_{eq} = 1/R_1 + 1/R_2 + 1/R_3$. For two resistors $R_{eq} = R_1 R_2 / (R_1 + R_2)$, and the current divides as $I_1 = I R_2 / (R_1 + R_2)$.

Series-parallel: reduce each parallel group first, then add the series resistances step by step until a single equivalent resistance is obtained.

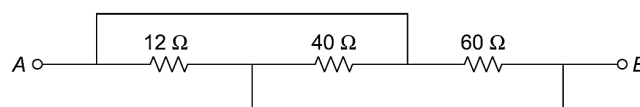


Figure 15: Series resistances.

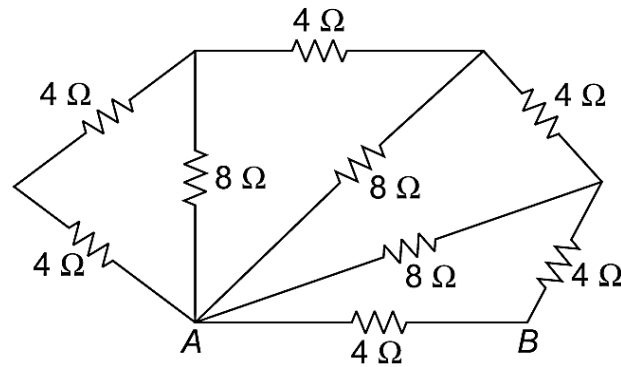


Figure 16: A series-parallel circuit.

Q3. State and explain Kirchhoff's laws with examples, and explain nodes, branches, loops and meshes.

Answer:

KCL: the algebraic sum of the currents at a junction is zero — currents entering equal currents leaving. Based on conservation of charge.

KVL: in any closed loop, the algebraic sum of emfs and voltage drops is zero, i.e. $+V_1 - V_2 - V_3 = 0$ around a loop. Based on conservation of energy.

- **Node:** a junction where two or more elements meet.
- **Branch:** a section between two junctions.
- **Loop:** any closed path.
- **Mesh:** a closed loop containing no other loop.

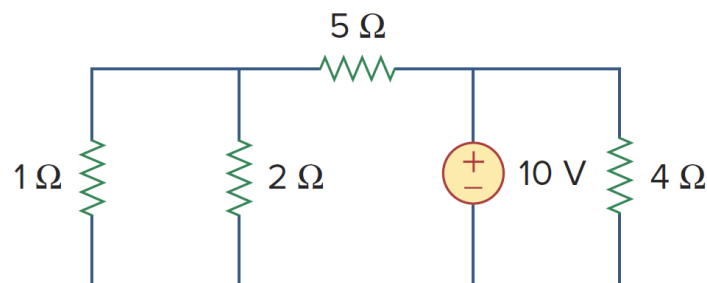


Figure 17: KVL around a loop.

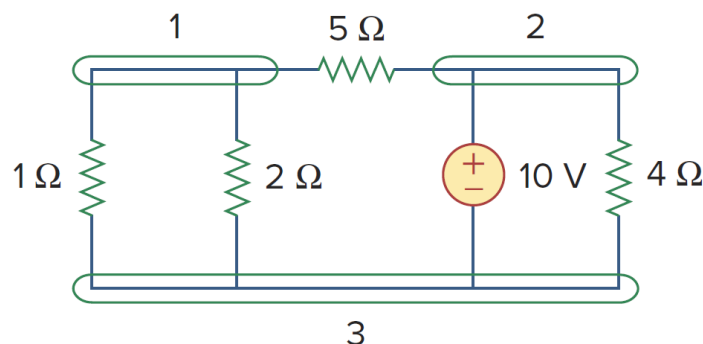


Figure 18: Nodes, branches, loops and meshes.

Q4. Explain star-delta and delta-star conversion in detail with formulas and a numerical example.

Answer:

A T (star/Y) network can be redrawn as an electrically equivalent delta (Δ) network and vice-versa. This is used when a network cannot be simplified by series and parallel combinations alone.

$$\Delta \rightarrow Y: R_1 = R_{AB}R_{CA}/(R_{AB}+R_{BC}+R_{CA}), R_2 = R_{AB}R_{BC}/(\text{sum}), R_3 = R_{BC}R_{CA}/(\text{sum})$$

$$Y \rightarrow \Delta: R_{AB} = (R_1R_2+R_2R_3+R_3R_1)/R_3, \text{ and similarly for } R_{BC} \text{ and } R_{CA}$$

Numerical: converting a given delta network to an equivalent star network gives star resistances such as $R_a = 140 \Omega$, $R_b = 70 \Omega$ and $R_c = 35 \Omega$.

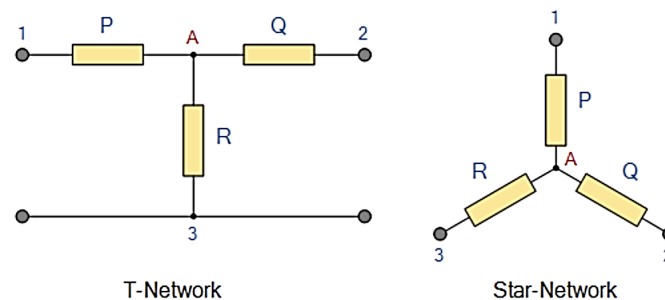


Figure 19: Delta network and its equivalent star network.

$$R_A = \frac{R_{AB} \cdot R_{CA}}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_B = \frac{R_{AB} \cdot R_{BC}}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_C = \frac{R_{BC} \cdot R_{CA}}{R_{AB} + R_{BC} + R_{CA}}$$

Figure 20: Worked delta-to-star example.

Q5. Explain mesh analysis in detail with the steps and a solved example.

Answer:

A mesh is a loop that contains no other loop. Mesh analysis is applicable only to planar networks and assigns a circulating current to each mesh; KVL is then written for each mesh in terms of these mesh currents. It is particularly useful when the network has many voltage sources.

- 1 Identify the meshes, assign a direction and an unknown mesh current to each.
- 2 Assign the polarities of the branch voltages.
- 3 Apply KVL to each mesh and express the branch voltages using Ohm's law.
- 4 Solve the simultaneous equations for the mesh currents.

The equations can be arranged as $[R][I] = [V]$, where the diagonal terms are the self-resistances of each mesh and the off-diagonal terms are the common (mutual) resistances.

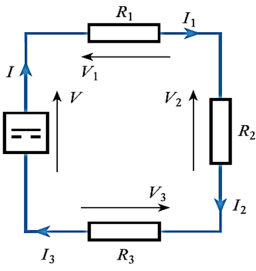
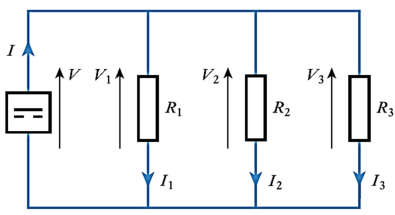
	Series circuit	Parallel network
		
Current	The current is the same in all parts of the circuit $I = I_1 = I_2 = I_3$	The total current supplied to the network equals the sum of the currents in the various branches $I = I_1 + I_2 + I_3$
Voltage	The total voltage equals the sum of the voltages across the different parts of the circuit $V = V_1 + V_2 + V_3$	The voltage across a parallel combination is the same as the voltage across each branch $V = V_1 = V_2 = V_3$
Resistance	The total resistance equals the sum of the separate resistances $R = R_1 + R_2 + R_3$	The reciprocal of the equivalent resistance equals the sum of the reciprocals of the branch resistances $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

Figure 21: A network solved by mesh analysis.

Q6. Explain nodal analysis in detail with the steps and a solved example.**Answer:**

Nodal analysis selects a reference node (ground) and writes Kirchhoff's current law at every other node in terms of the node voltages. It is particularly useful when the network contains many current sources.

- 1 Choose a reference node and label the remaining node voltages $V_1, V_2, \dots$ with respect to it.
- 2 Assume all currents to leave the node under consideration.
- 3 Apply KCL at each non-reference node, writing each branch current as (node-voltage difference)/resistance.
- 4 Solve the simultaneous equations for the node voltages.

For example, the current leaving node A towards node B through resistance R is $(V_A - V_B)/R$. Once the node voltages are known, any branch current follows from Ohm's law.

Q7. Explain source transformation and the superposition theorem in detail.**Answer:**

Source transformation: a voltage source with a series resistance can be replaced by an equivalent current source with a parallel resistance, and vice-versa. The relations are $I_S = V_S/R_S$ and $R_P = R_S$.

Superposition theorem: in a linear, bilateral network with several sources, the response in any element equals the algebraic sum of the responses caused by each source acting alone, with all other voltage sources short-circuited and current sources open-circuited.

- 1 Select one source and suppress the others (voltage sources shorted, current sources opened).
- 2 Find the desired voltage or current due to that source.
- 3 Repeat for each remaining source.

4 Add all the contributions algebraically.

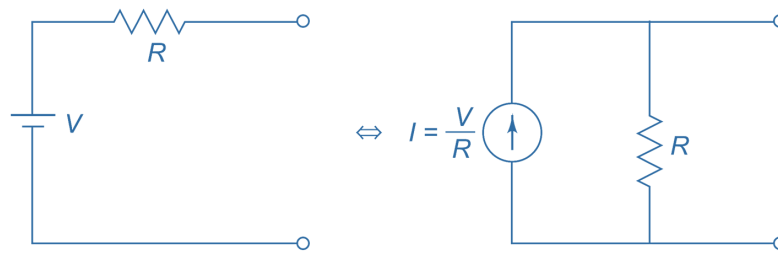


Figure 22: Source transformation.

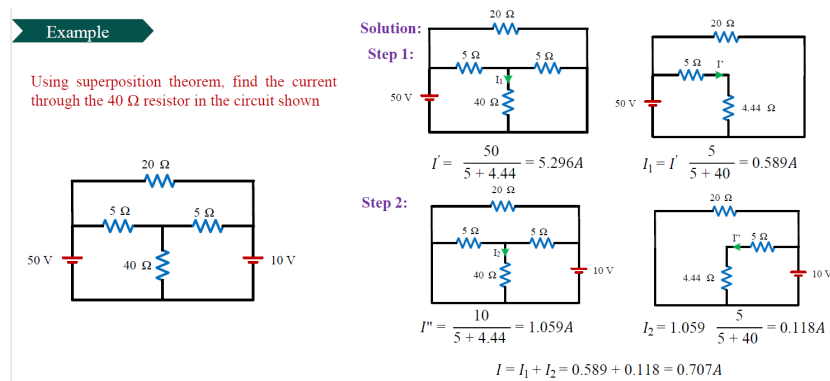


Figure 23: Superposition — worked example.

Q8. State and explain Thevenin's and Norton's theorems with the procedures and a solved example.

Answer:

Thevenin's theorem: any linear bilateral network with terminals A-B can be replaced by a voltage source V_{Th} in series with a resistance R_{Th} . V_{Th} is the open-circuit voltage across A-B; R_{Th} is the resistance seen from A-B with voltage sources shorted and current sources opened.

Norton's theorem: the same network can be replaced by a current source I_N in parallel with R_N , where I_N is the short-circuit current and $R_N = R_{Th}$.

Solved example: for a network with a 4.5 V and a 3 V source, $V_{Th} = 3.83 V$ and $R_{Th} = 2.22 \Omega$, giving a load current $I = 3.83 / (2.22 + 6) = 0.47 A$ through a 6 Ω load.

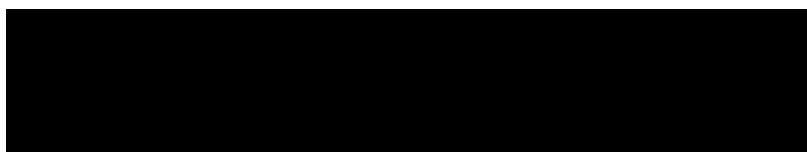


Figure 24: Thevenin's theorem.

Procedure for Finding Thevenin Equivalent Circuit

1. Open the two terminals (i.e., remove any load) between which you want to find Thevenin equivalent circuit.
2. Find the open-circuit voltage between the two open terminals. It is called Thevenin voltage V_{Th} .
3. Determine the resistance between the two open terminals with all voltage sources shorted and all current sources opened. It is called Thevenin resistance R_{Th} .
4. Connect V_{Th} and R_{Th} in series to produce Thevenin equivalent circuit between the two terminals under consideration.
5. Place the load resistor removed in step (1) across the terminals of the Thevenin equivalent circuit. The load current can now be calculated using only Ohm's law and it has the same value as the load current in the original circuit.

Figure 25: Worked Thevenin example.

Q9. State and prove the maximum power transfer theorem, and solve a numerical example.

Answer:

Statement: maximum power is transferred from a source to the load when the load resistance equals the internal resistance of the source as viewed from the load terminals ($R_L = R_i$).

Proof: the load power is $P_L = I^2 R_L = [V/(R_L + R_i)]^2 R_L$. Differentiating with respect to R_L and equating dP_L/dR_L to zero gives $(R_L + R_i) - 2R_L = 0$, i.e. $R_L = R_i$.

Important points: at maximum power transfer the efficiency is only 50%, the load voltage is half the open-circuit voltage, and $P_{max} = V^2/(4R_L)$.

Numerical: for a 120 V source with 6 A source and 10 Ω , 5 Ω resistors, $V_{Th} = 60$ V and $R_{Th} = 3.33$ Ω , so $R = 3.33$ Ω and $P_{max} = 60^2/(4 \times 3.33) = 270$ W.

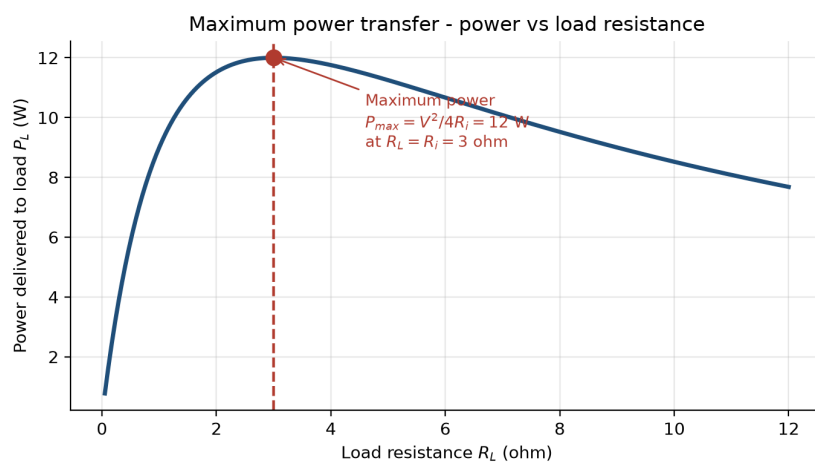


Figure 26: Power vs load resistance.

In d.c. circuits, maximum power is transferred from a source to load when the load resistance is made equal to the internal resistance of the source as viewed from the load terminals with load removed and all e.m.f. sources replaced by their internal resistances.

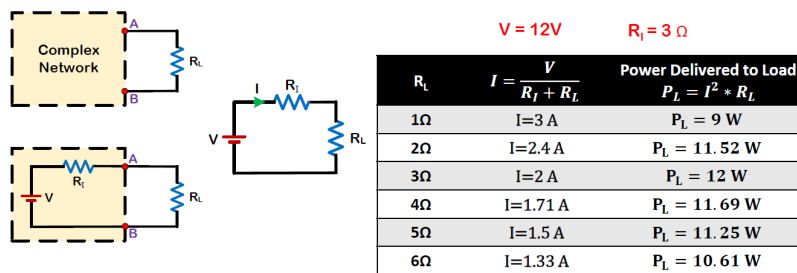


Figure 27: Worked maximum-power example.

SECTION D — Numerical Problems (Solved)

Every numerical problem below is solved completely, step by step, and the circuit diagram is given with each problem. Read the method carefully — the same steps work for the exam problems.

N1. Series Circuit

Problem: A 12 V source is connected to $R_1 = 2\Omega$ and $R_2 = 4\Omega$ in series. Find the current, the voltage across each resistor and the total power.

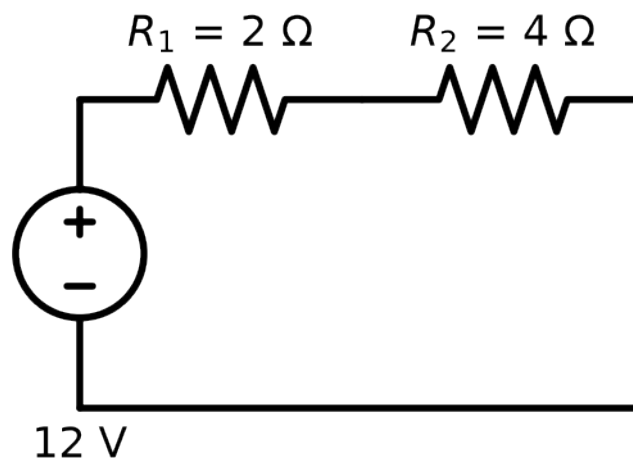


Figure 28: Circuit for N1.

Solution:

Step 1 — equivalent resistance (series): $R_{eq} = R_1 + R_2 = 2 + 4 = 6\Omega$.

Step 2 — current (Ohm's law): $I = V / R_{eq} = 12 / 6 = 2\text{ A}$. The same current flows through both resistors.

Step 3 — voltages (voltage division): $V_1 = I R_1 = 2 \times 2 = 4\text{ V}$; $V_2 = I R_2 = 2 \times 4 = 8\text{ V}$ (and $4 + 8 = 12\text{ V}$ — checks).

Step 4 — total power: $P = V I = 12 \times 2 = 24\text{ W}$.

Answer: $I = 2\text{ A}$, $V_1 = 4\text{ V}$, $V_2 = 8\text{ V}$, $P = 24\text{ W}$.

N2. Parallel Circuit

Problem: A 12 V source is connected to $R_1 = 6\ \Omega$ and $R_2 = 3\ \Omega$ in parallel. Find the equivalent resistance, the branch currents and the total current.

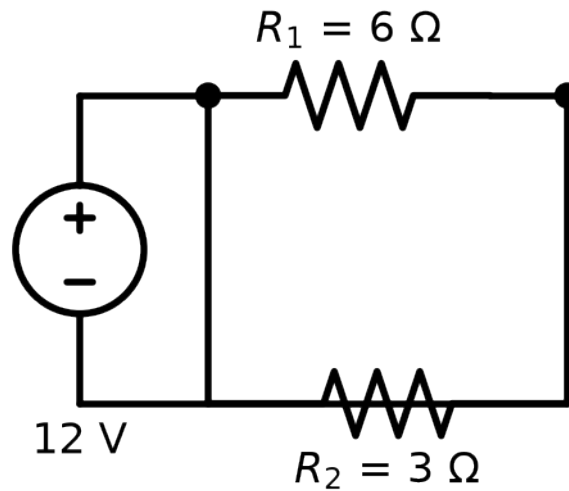


Figure 29: Circuit for N2.

Solution:

Step 1 — equivalent resistance (parallel): $R_{eq} = R_1 R_2 / (R_1 + R_2) = (6 \times 3) / (6 + 3) = 18/9 = 2\ \Omega$.

Step 2 — branch currents (the same 12 V appears across each branch): $I_1 = 12/6 = 2\ \text{A}$ and $I_2 = 12/3 = 4\ \text{A}$.

Step 3 — total current: $I = I_1 + I_2 = 6\ \text{A}$ (also $I = V/R_{eq} = 12/2 = 6\ \text{A}$ — checks).

Answer: $R_{eq} = 2\ \Omega$, $I_1 = 2\ \text{A}$, $I_2 = 4\ \text{A}$, $I = 6\ \text{A}$.

N3. Series-Parallel Circuit

Problem: A 24 V source feeds $R_1 = 4\ \Omega$ in series with a parallel combination of $R_2 = 6\ \Omega$ and $R_3 = 3\ \Omega$. Find the total current and the current in each parallel branch.

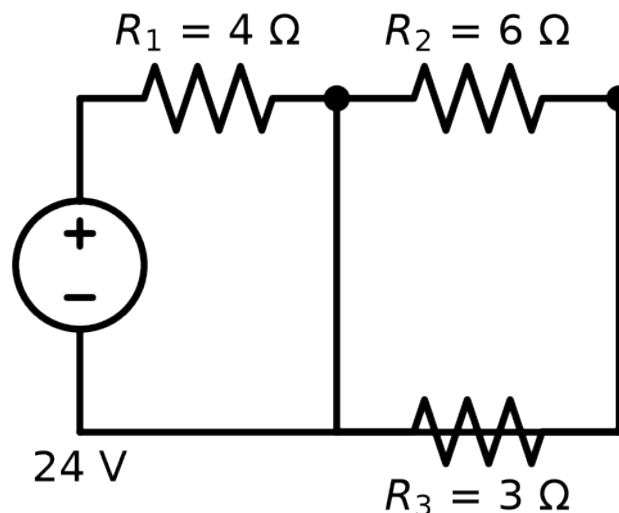


Figure 30: Circuit for N3.

Solution:

Step 1 — reduce the parallel pair: $R_{23} = (6 \times 3) / (6 + 3) = 2\ \Omega$.

Step 2 — total resistance: $R_{eq} = R_1 + R_{23} = 4 + 2 = 6\ \Omega$.

Step 3 — total current: $I = 24 / 6 = 4 \text{ A}$.

Step 4 — voltage across the parallel part: $V_{23} = I R_{23} = 4 \times 2 = 8 \text{ V}$.

Step 5 — branch currents: $I_2 = 8/6 = 1.33 \text{ A}$ and $I_3 = 8/3 = 2.67 \text{ A}$ (and $1.33 + 2.67 = 4 \text{ A}$ — checks).

Answer: $I = 4 \text{ A}$, $I_2 = 1.33 \text{ A}$, $I_3 = 2.67 \text{ A}$.

N4. Kirchhoff's Current Law (KCL)

Problem: At a node, a current $I_1 = 4 \text{ A}$ and a current $I_2 = 3 \text{ A}$ enter the node, and a current I_3 leaves it. Find I_3 .

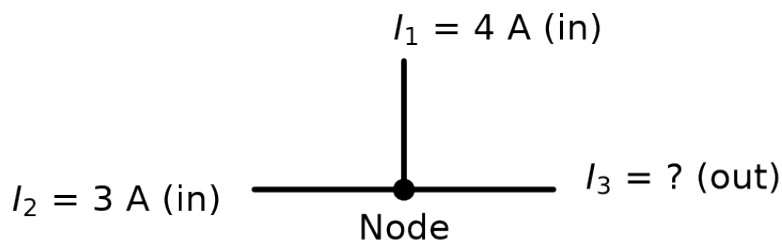


Figure 31: Circuit for N4.

Solution:

KCL states that the sum of the currents entering a node equals the sum of the currents leaving it.

Sum entering $= I_1 + I_2 = 4 + 3 = 7 \text{ A}$.

Therefore the current leaving, $I_3 = 7 \text{ A}$.

Answer: $I_3 = 7 \text{ A}$.

N5. Kirchhoff's Voltage Law (KVL)

Problem: A single loop contains a 12 V source, a 4 V source opposing it, and resistors $R_1 = 2 \Omega$ and $R_2 = 3 \Omega$. Find the loop current.

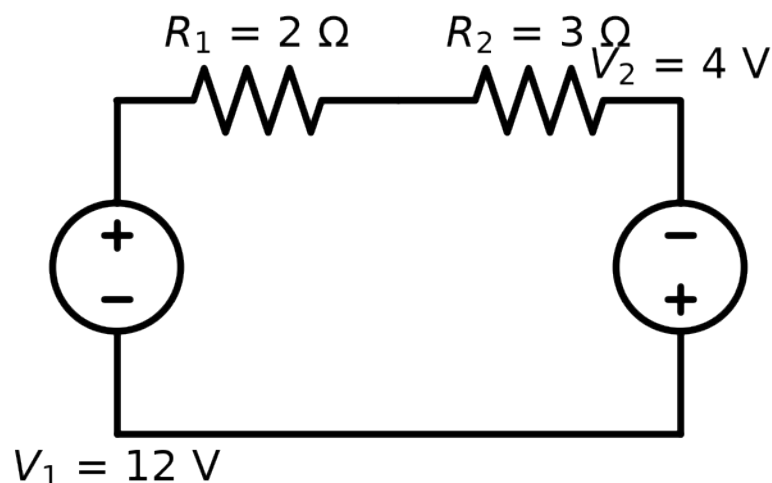


Figure 32: Circuit for N5.

Solution:

Step 1 — net emf in the loop: the two sources oppose, so $V_{\text{net}} = 12 - 4 = 8 \text{ V}$.

Step 2 — total resistance: $R = R_1 + R_2 = 2 + 3 = 5 \Omega$.

Step 3 — by KVL and Ohm's law: $I = V_{\text{net}} / R = 8 / 5 = 1.6 \text{ A}$.

Answer: $I = 1.6 \text{ A}$.

N6. Mesh Analysis

Problem: Using mesh analysis, find the mesh currents and the current through the 4Ω resistor in the two-mesh network shown (10 V source; $R_1 = 5 \Omega$, $R_2 = 3 \Omega$, $R_3 = 1 \Omega$ and $R_4 = 4 \Omega$).

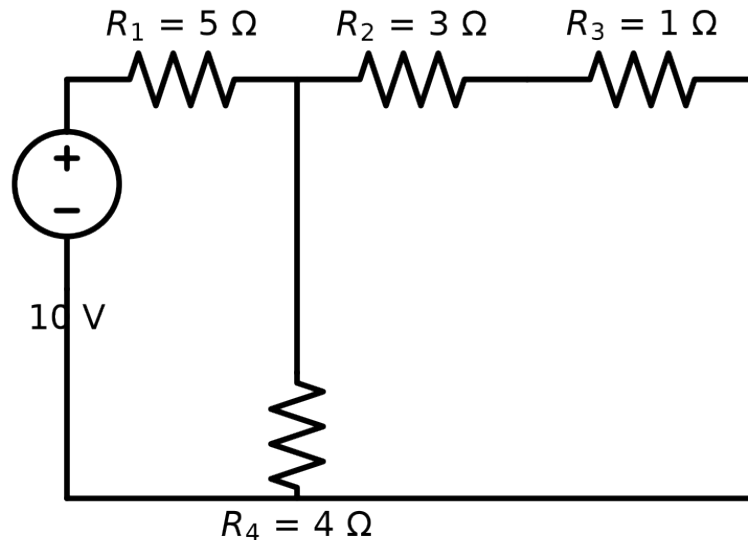


Figure 33: Circuit for N6.

Solution:

Step 1 — assign clockwise mesh currents i_1 (left mesh) and i_2 (right mesh).

Step 2 — KVL in mesh 1: $10 - 5 i_1 - 4(i_1 - i_2) = 0 \Rightarrow 9 i_1 - 4 i_2 = 10$.

Step 3 — KVL in mesh 2: $-3 i_2 - 1 i_2 - 4(i_2 - i_1) = 0 \Rightarrow 4 i_1 - 8 i_2 = 0 \Rightarrow i_1 = 2 i_2$.

Step 4 — substituting: $9(2 i_2) - 4 i_2 = 10 \Rightarrow 14 i_2 = 10 \Rightarrow i_2 = 0.71 \text{ A}$, and $i_1 = 1.43 \text{ A}$.

Step 5 — the current through the 4Ω resistor is $(i_1 - i_2) = 0.71 \text{ A}$.

Answer: $i_1 = 1.43 \text{ A}$, $i_2 = 0.71 \text{ A}$, current through $4 \Omega = 0.71 \text{ A}$.

N7. Nodal Analysis

Problem: Using nodal analysis, find the node voltages V_A and V_B in the circuit shown (a 6 A source feeds node A; $R_1 = 10 \Omega$ from A to ground; $R_2 = 5 \Omega$ between A and B; a 2 A source leaves node B).

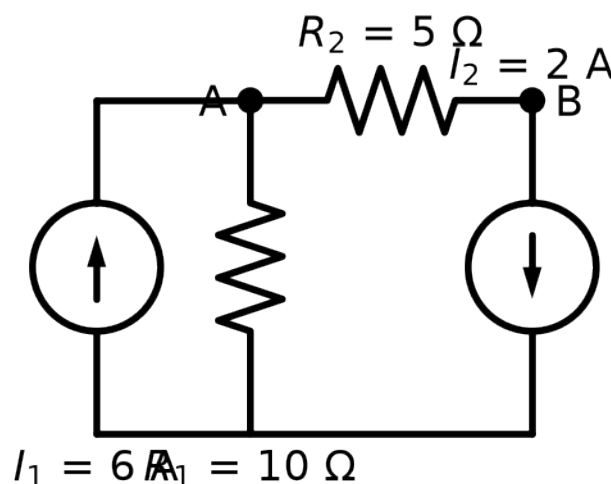


Figure 34: Circuit for N7.

Solution:

Step 1 — KCL at node B: the current through R_2 equals the current leaving through the 2 A source:
 $(V_A - V_B)/5 = 2 \Rightarrow V_A - V_B = 10$.

Step 2 — KCL at node A: $6 = V_A/10 + (V_A - V_B)/5$.

Step 3 — substituting $(V_A - V_B) = 10$: $6 = V_A/10 + 2 \Rightarrow V_A/10 = 4 \Rightarrow V_A = 40$ V.

Step 4 — hence $V_B = V_A - 10 = 30$ V. Check: $I_{R1} = 4$ A, $I_{R2} = 2$ A, total 6 A — correct.

Answer: $V_A = 40$ V, $V_B = 30$ V.

N8. Supermesh Analysis

Problem: In the network shown (8 V source with a 2 Ω resistor in the left mesh; a 2 A current source common to both meshes; a 6 Ω resistor in the right mesh), find the mesh currents and the current through the 6 Ω resistor.

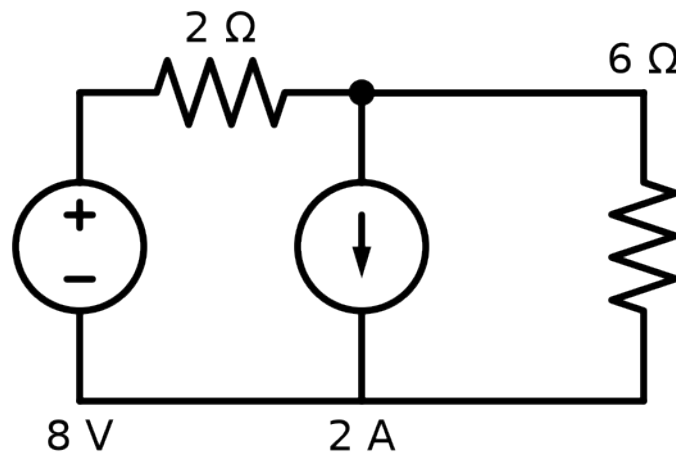


Figure 35: Circuit for N8.

Solution:

Step 1 — the current source is common to both meshes, so form a supermesh around the two meshes.

Step 2 — constraint from the current source: $i_1 - i_2 = 2$ A.

Step 3 — KVL around the supermesh: $8 - 2i_1 - 6i_2 = 0$.

Step 4 — substituting $i_1 = i_2 + 2$: $8 = 2(i_2 + 2) + 6i_2 = 8i_2 + 4 \Rightarrow i_2 = 0.5$ A and $i_1 = 2.5$ A.

Step 5 — the current through the 6 Ω resistor is $i_2 = 0.5$ A.

Answer: $i_1 = 2.5$ A, $i_2 = 0.5$ A, current through 6 $\Omega = 0.5$ A.

N9. Source Transformation

Problem: A 12 V voltage source has a 6 Ω resistor in series. Convert it into an equivalent current source.

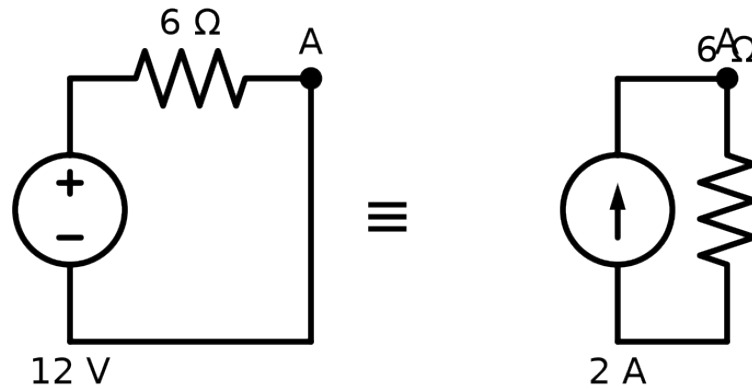


Figure 36: Circuit for N9.

Solution:

Step 1 — the equivalent current source value: $I_s = V_s / R_s = 12 / 6 = 2 \text{ A}$.

Step 2 — the resistor stays the same value but is now connected in parallel: $R_p = R_s = 6 \Omega$.

So the voltage source (12 V in series with 6 Ω) becomes a current source of 2 A in parallel with 6 Ω.

Answer: $I_s = 2 \text{ A}$ in parallel with 6 Ω.

N10. Superposition Theorem

Problem: Using the superposition theorem, find the current through the 10 Ω resistor in the network shown ($V_1 = 40 \text{ V}$ with a 10 Ω resistor; $V_2 = 20 \text{ V}$ with a 20 Ω resistor; a 10 Ω resistor between node N and the return path).

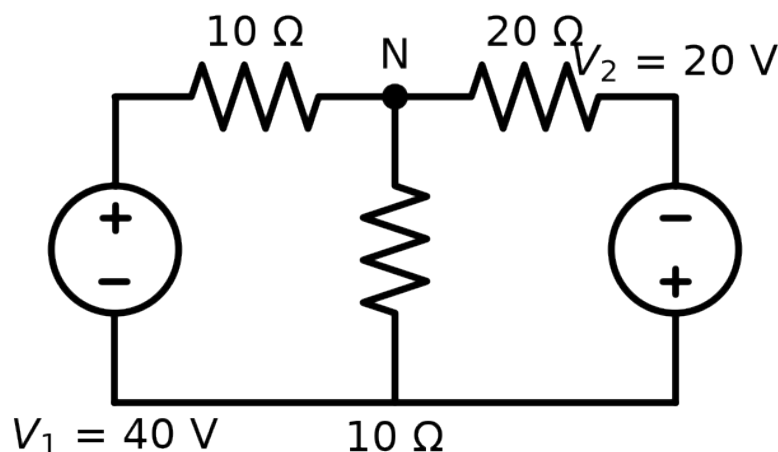


Figure 37: Circuit for N10.

Solution:

Step 1 — effect of V_1 alone (V_2 shorted): the 10 Ω and 20 Ω are in parallel = 6.67 Ω; total = 10 + 6.67 = 16.67 Ω; total current = $40/16.67 = 2.4 \text{ A}$. Current into the 10 Ω branch = $2.4 \times 20/30 = 1.6 \text{ A}$.

Step 2 — effect of V_2 alone (V_1 shorted): the 10 Ω and 10 Ω are in parallel = 5 Ω; total = 20 + 5 = 25 Ω; total current = $20/25 = 0.8 \text{ A}$. Current into the 10 Ω branch = $0.8 \times 10/20 = 0.4 \text{ A}$.

Step 3 — both contributions flow in the same direction through the 10 Ω resistor, so the total current = $1.6 + 0.4 = 2.0 \text{ A}$.

Check by nodal analysis: at node N, $(V_N - 40)/10 + V_N/10 + (V_N - 20)/20 = 0 \Rightarrow 5V_N = 100 \Rightarrow V_N = 20 \text{ V}$, so current = $20/10 = 2 \text{ A}$ — correct.

Answer: Current through the $10\ \Omega$ resistor = 2 A.

N11. Thevenin's Theorem

Problem: Using Thevenin's theorem, find the current through the load $R_L = 6\ \Omega$ in the circuit shown (24 V source; a $7\ \Omega$ resistor in series; a $10\ \Omega$ resistor across the terminals; a $4\ \Omega$ resistor in series with the load).

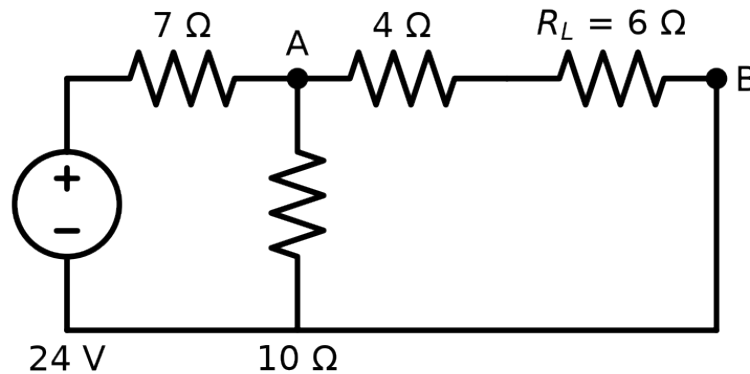


Figure 38: Circuit for N11.

Solution:

Step 1 — remove the load R_L . Find the open-circuit (Thevenin) voltage across A-B. With R_L removed, no current flows in the $4\ \Omega$ resistor, so $V_{Th} =$ voltage across the $10\ \Omega$ resistor = $24 \times 10/(7 + 10) = 14.12\text{ V}$.

Step 2 — find R_{Th} with the source shorted: $R_{Th} = (7 \parallel 10) + 4 = (7 \times 10/17) + 4 = 4.12 + 4 = 8.12\ \Omega$.

Step 3 — reconnect the load and find the load current: $I_L = V_{Th} / (R_{Th} + R_L) = 14.12 / (8.12 + 6) = 1.0\text{ A}$.

Answer: $V_{Th} = 14.12\text{ V}$, $R_{Th} = 8.12\ \Omega$, $I_L = 1\text{ A}$.

N12. Maximum Power Transfer Theorem

Problem: A source of $V = 12\text{ V}$ has an internal resistance $R_i = 3\ \Omega$. Find the value of the load resistance R_L for maximum power transfer, and the maximum power delivered to the load.

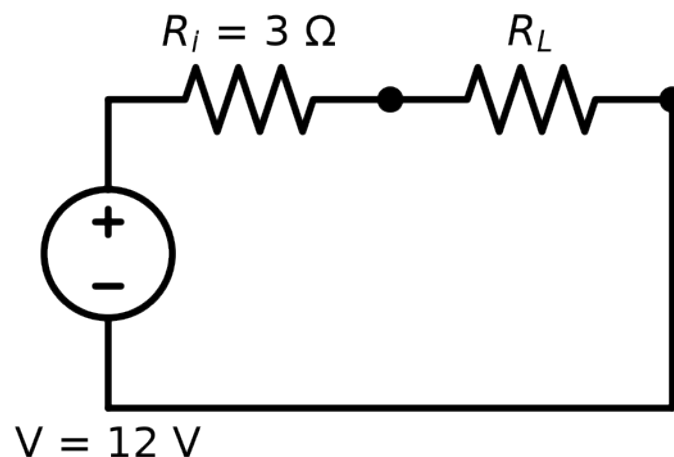


Figure 39: Circuit for N12.

Solution:

Step 1 — condition for maximum power transfer: $R_L = R_i = 3\ \Omega$.

Step 2 — current at this condition: $I = V/(R_L + R_i) = 12/(3 + 3) = 2 \text{ A}$.

Step 3 — maximum power: $P_{\max} = I^2 R_L = 2^2 \times 3 = 12 \text{ W}$ (also $V^2/(4R_L) = 144/12 = 12 \text{ W}$).

Note: the efficiency at maximum power transfer is only 50% — half the power is lost in R_i .

Answer: $R_L = 3 \Omega$, $P_{\max} = 12 \text{ W}$.